

Lecture 4

Descent and mirror symmetry

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Research seminar notes

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1 Covariant wrapped Fukaya descent and its applications

1.1 The descent theorem

Let (X, λ) be a complete finite-type Weinstein manifold, with $\omega = d\lambda$ and Liouville vector field $\iota_Z \omega = \lambda$. Let $K_1, \dots, K_N \subset X$ be Poisson-commuting compact codimension-zero submanifolds with corners whose boundary faces meet transversely. Assume that the Liouville vector field Z points strictly outward along every boundary face of every K_i , and that

$$\mathrm{Sk}(Z) \subseteq \bigcup_i \mathrm{int}(K_i).$$

For a nonempty index set I , put $K_I = \bigcap_{i \in I} K_i$. Smooth the corners of K_I in a small neighbourhood of its boundary, keeping Z strictly outward, and attach the positive symplectization collar to the resulting smooth boundary. Write the completion as $\widehat{K}_I$. To see that the choice of smoothing does not matter, use a flow box for Z : the relevant boundaries are graphs transverse to the flow, and moving one graph to another along its flow lines identifies their completions. Empty intersections are assigned the zero category. We write Mod for derived unital right A_∞ -modules; all functors and compatible collections below are derived and homotopy coherent.

Theorem 1.1 (Wrapped descent). *Viterbo restriction gives equivalences of stable ∞ -categories*

$$\begin{aligned} \mathrm{Perf} \mathcal{W}(X) &\simeq \mathrm{holim}_{\emptyset \neq I \subseteq \{1, \dots, N\}} \mathrm{Perf} \mathcal{W}(\widehat{K}_I), \\ \mathrm{Mod} \mathcal{W}(X) &\simeq \mathrm{holim}_{\emptyset \neq I \subseteq \{1, \dots, N\}} \mathrm{Mod} \mathcal{W}(\widehat{K}_I). \end{aligned}$$

The diagrams on the right use Viterbo restriction, in the framework of Ganatra–Pardon–Shende and Sylvan [24, Section 11.1][2].

The comparison with the supported category from Lecture 3 is the following. For exact Lagrangians L_0, L_1 which are conical near ∂K_I and at infinity, with primitives vanishing on these collars, put $L_j^I = \widehat{L_j} \cap \widehat{K}_I$. Then

$$HF_X^*(L_0, L_1; K_I; \mathbb{Z}) \cong HW_{\widehat{K}_I}^*(L_0^I, L_1^I; \mathbb{Z}).$$

These identifications are compatible with products and restriction of support. This is the support–wrapped comparison of Lecture 3, Section 6.1; its chain-level version identifies the corresponding morphism complexes. The descent theorem for those complexes is the input from [1].

Cieliebak’s weaker condition. The global Weinstein assumption on λ can be replaced by the following local assumption on the *induced* structures on the $\widehat{K}_i$. Each admits an exhausting function ϕ_i and a smooth positive bilinear form g_i such that

$$d\phi_i = g_i(Z, \cdot), \quad g_i(v, v) > 0 \quad (v \neq 0).$$

The tensor g_i need not be symmetric, and no Morse or Morse–Bott condition is imposed on the critical locus. This is Cieliebak’s gradient-like condition [3, Equation (3), Lemma 6]. We retain the common outward field, the support-covering condition, and finite-type completeness of X .

Here is why this local condition suffices. Choose slightly smaller flow roundings whose interiors still cover the skeleton. On each, adapt ϕ_i near the boundary to have constant regular maximum, without altering it near $Z = 0$. Compose it with an increasing function flat at

that maximum, subtract the boundary value, and extend by zero. Here flat means vanishing to infinite order at the boundary value, so the functions and their associated tensors extend smoothly. The sum ψ of these functions satisfies

$$d\psi = g(Z, \cdot)$$

with g a sum of positive-semidefinite tensors, positive wherever at least one summand is interior. A negative regular sublevel of ψ contains the skeleton and is a full Liouville domain. Extend ψ over its end. Cieliebak's perturbation lemma now makes the function Morse by a small exact change $\lambda \mapsto \lambda + dF$, supported near $Z = 0$. This support can avoid all the original boundary faces: strict outwardness implies that none contains a zero of Z . The symplectic form and the Poisson-commuting supports are unchanged, and the Liouville field is unchanged near every face. The theorem therefore applies after this perturbation, and deformation invariance identifies its restriction diagram with the original one.

1.2 Hacking–Keating HMS by induction

An exact ray diagram R consists of finitely many rational rays from the origin, with a positive multiplicity on each ray. Its almost-toric total space will be denoted by U_R . A multiplicity n means that the ray carries n focus–focus points. These are the surface models discussed in Lecture 2.

On the regular locus, use action–angle coordinates (x_j, θ_j) centred at the origin. The standard primitive and its Liouville field are

$$\lambda = \sum_{j=1}^2 x_j d\theta_j, \quad Z = \sum_{j=1}^2 x_j \frac{\partial}{\partial x_j}.$$

Thus Z is the horizontal lift of the Euler vector field on the base. The shears across the eigenrays preserve this field; extend its lift across small neighbourhoods of the nodal fibres. These choices give the same complete Liouville manifold up to strongly exact symplectomorphism. We use the unique grading structure for which the regular torus fibres have Maslov class zero, and equip the Lagrangian objects with the spin and local-system data of Lecture 2. See [25, Section 1.1.1].

The mirror X_R has an elementary algebraic description. Choose a smooth complete toric surface over $\mathbb{Z}$ whose fan contains the given rays. At the point -1 of the open toric boundary component corresponding to each ray, perform the prescribed number of interior blowups. For repeated blowups, the next centre is the intersection of the last exceptional curve with the proper transform of the boundary. Finally remove the proper transform of the toric boundary. Extra toric subdivisions do not affect the resulting open surface. Repeated blowups can leave compact exceptional curves, so X_R need not be affine.

Hacking–Keating prove HMS for the distinguished complex structures on log Calabi–Yau surfaces, and identify their symplectic mirrors, up to Weinstein deformation, with the almost-toric models [4]. Over $k = \mathbb{C}$, their conclusion in this notation is

$$D^\pi \mathcal{W}(U_R; \mathbb{C}) \simeq D^b \text{Coh}((X_R)_{\mathbb{C}}) = \text{Perf}((X_R)_{\mathbb{C}}).$$

Their proof uses Lefschetz fibrations. The descent argument below works with integral coefficients: its local algebra is $\mathbb{Z}[x, y, u^{\pm 1}]/(xy - 1 - u)$ and its algebraic mirror is the scheme X_R over $\mathbb{Z}$. It gives

$$\text{Perf } \mathcal{W}(U_R; \mathbb{Z}) \simeq \text{Perf}(X_R).$$

Base change to a field gives the corresponding field-valued statement.

The two basic pieces. For the empty diagram the mirror pair is

$$T^*T^2 \longleftrightarrow (\mathbb{G}_m)^2.$$

For a single node, write U_1 for the symplectic model. Its mirror is

$$X_1 = \operatorname{Spec} R_1, \quad R_1 = \mathbb{Z}[x, y, u^{\pm 1}]/(xy - 1 - u).$$

A section gives a generator with degree-zero endomorphism algebra R_1 ; the elementary product has two contributions and gives $xy = 1 + u$. Thus $\operatorname{Perf} \mathcal{W}(U_1) \simeq \operatorname{Perf}(R_1)$. The local model is self-mirror; it is also the case $X_{1,1}$ of [9]. The homological-section generation criterion gives another route to this description over a characteristic-zero field [10]. The two torus charts are visible without any further calculation:

$$R_1[x^{-1}] = \mathbb{Z}[x^{\pm 1}, u^{\pm 1}], \quad R_1[y^{-1}] = \mathbb{Z}[y^{\pm 1}, u^{\pm 1}], \quad y = (1 + u)/x.$$

These two torus charts miss the closed section $V(x, y, u + 1) \simeq \operatorname{Spec} \mathbb{Z}$, which is retained in the nodal chart. The matching A-side restriction is the Viterbo map for removing the appropriate visible tail. This identification of the restriction functor is part of the local input to the induction.

Peeling off a node. First prepare a ray carrying exactly one node. Insert the opposite ray into the toric fan by corner blowups if necessary; these preserve the open surface. If the chosen ray has $m > 1$ nodes, move its innermost node through the origin and transfer the cut; repeat $m - 1$ times. At each step place the moved node between the origin and any nodes already on the opposite ray. This involves no crossing of nodes. The accompanying piecewise integral-affine shear changes the other ray directions but preserves the number of nodes. On the algebraic side these are elementary transformations of the toric model, which preserve the open surface. This is the nodal-slide/cut-transfer operation of [4, Definition 3.24, Lemma 3.25 and Section 6.1.3]. It changes the multiplicities by $m_v \mapsto m_v - 1$ and $m_{-v} \mapsto m_{-v} + 1$. We denote the resulting presentation again by R . Let b_* be its single node on the chosen ray and let R' be obtained by deleting that node.

Take two compact regions P_0, P_1 in the base, both containing a small neighbourhood of the origin and strictly star-shaped for the Euler field. Let P_1 extend along the chosen ray past b_* and contain no other node. Let P_0 extend along all the remaining rays past all their nodes, but stop before b_* on the chosen ray. Arrange that their intersection lies before the start of every eigenray, in the flat reference chart, and their union contains the radial segments joining the origin to all nodes (Figure 1). The regions are thickened slightly so that their boundaries are transverse; they may also be enlarged along the ends without changing these properties. An occupied opposite ray causes no difficulty: it belongs to P_0 .

Put $K_i = \pi^{-1}(P_i)$. Choose defining functions on the base which are constant near the nodes and pull them back by π . They Poisson commute: in the regular action-angle charts they depend only on the actions. The common Liouville field is the radial lift near all boundary faces, so it points strictly outward there. Use the nodal extensions satisfying Cieliebak's condition from Section 1.1 in the interiors. The same primitive then works for both pieces, their intersection and their union. After rounding the corners, $K_0 \cup K_1$ is a full Liouville domain, and

$$\widehat{K}_0 \simeq U_{R'}, \quad \widehat{K}_1 \simeq U_1, \quad \widehat{K}_{01} \simeq T^*T^2.$$

Indeed, radial completion extends the outer boundary to infinity and retains exactly the nodes lying in each region.

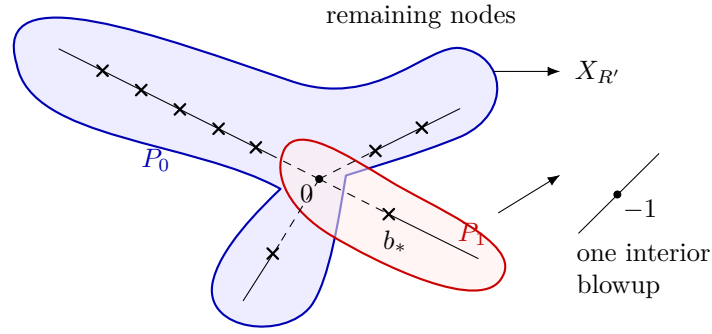


Figure 1: The two-piece cover after preparing a ray with one node. The overlap contains no node; its completion is T^*T^2 . On the mirror, the added node is one blowup at -1 on an open toric boundary component.

On the mirror, let E_* be the exceptional curve of the single blowup on the boundary component D_v . Deleting E_* gives $V_0 \simeq X_{R'}$. The toric open subset $T \cup D_v^\circ \simeq \mathbb{A}^1 \times \mathbb{G}_m$ contains no other blowup centre, so its transformed complement of the proper boundary is $V_1 \simeq X_1$. Hence there is an open cover

$$X_R = V_0 \cup V_1, \quad V_0 \simeq X_{R'}, \quad V_1 \simeq X_1, \quad V_0 \cap V_1 \simeq (\mathbb{G}_m)^2.$$

The overlap is the complement of the visible tail in the nodal piece; its coordinate ring is $R_1[x^{-1}]$ (or $R_1[y^{-1}]$ for the opposite choice of tail). Thus the one-node equivalence identifies Viterbo restriction with this localization. On the other piece it is the restriction to the same exposed torus chart.

Both sides are now computed by the same homotopy fibre product:

$$\begin{aligned} \mathrm{Perf} \mathcal{W}(U_R) &\simeq \mathrm{Perf} \mathcal{W}(U_{R'}) \underset{\mathrm{Perf} \mathcal{W}(T^*T^2)}{\overset{h}{\times}} \mathrm{Perf} \mathcal{W}(U_1), \\ \mathrm{Perf}(X_R) &\simeq \mathrm{Perf}(X_{R'}) \underset{\mathrm{Perf}((\mathbb{G}_m)^2)}{\overset{h}{\times}} \mathrm{Perf}(X_1). \end{aligned}$$

The first equivalence is Theorem 1.1; the second is Zariski descent. The induction hypothesis and the one-node equivalence identify the three entries and their restriction maps, and therefore identify their homotopy limits. Starting from zero nodes proves the result.

Exercise. Identify the mirrors of the following objects: the reference section, a terminal visible tail, and a Lagrangian sphere over the segment between two successive nodes on the same ray. Show that they are respectively $\mathcal{O}_{X_R}$, the structure sheaf of the corresponding open exceptional curve, and $\mathcal{O}_{\mathbb{P}^1}(-1)$ on the corresponding compact exceptional curve, pushed forward to X_R . Choose the gradings so that these sheaves are in degree zero. You may use the local one-node calculation and the two-piece descent diagram.

1.3 The expected mirror of $\mathbb{P}^3$ minus a quartic K3

Let $Q \subset \mathbb{P}^3$ be a smooth quartic and set $U = \mathbb{P}^3 \setminus Q$. We use the SYZ model from Lecture 2, with four positive and twenty-four negative vertices (Figure 2). For this application we assume the weakened Cieliebak–Weinstein hypothesis in the descent theorem. We now describe the expected algebraic mirror.

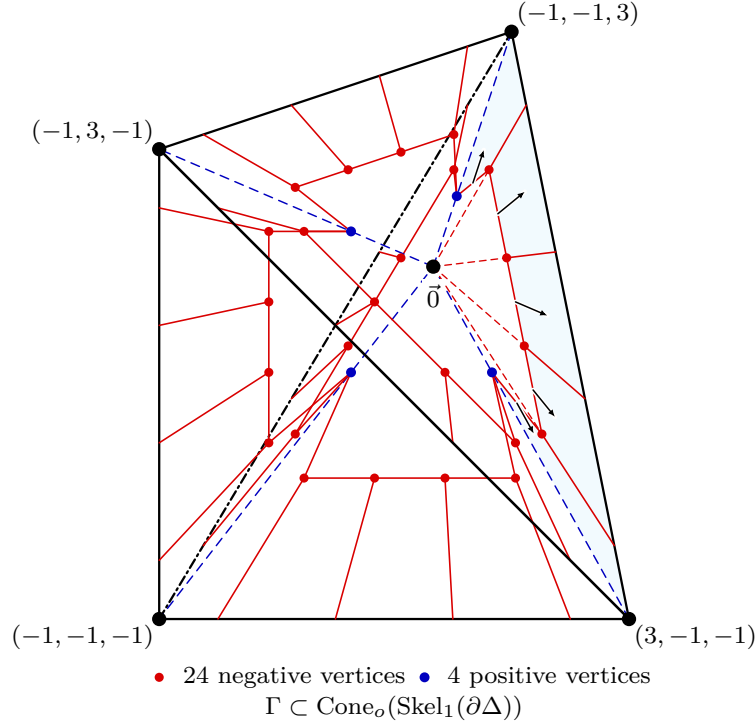


Figure 2: The tetrahedral model from Lecture 2. Each of the six edge packages has four negative vertices (red); the four positive vertices are blue. The shaded branch sheet, radial subdivisions, and monodromy vectors are retained from the final version of that figure. The tetrahedral edges have integral length four.

The toric blowup construction. Let

$$\Delta = \text{Conv}\{(-1, -1, -1), (3, -1, -1), (-1, 3, -1), (-1, -1, 3)\} \subset N_{\mathbb{R}},$$

and choose a smooth projective subdivision of the fan over its faces, with rays through all boundary lattice points. Write V_{Σ} for the resulting toric threefold and D_{Σ} for its toric boundary. The polar polytope is

$$\Delta^{\vee} = \text{Conv}\{e_1, e_2, e_3, -e_1 - e_2 - e_3\}.$$

On the dense torus, consider the Laurent polynomial

$$W = x + y + z + \frac{1}{xyz}.$$

If s_{∞} is the toric anticanonical section with divisor D_{Σ} , then $s_0 = Ws_{\infty}$ extends to an anticanonical section. In Cox coordinates one can write it explicitly as

$$s_0 = \sum_{m \in \{e_1, e_2, e_3, -e_1 - e_2 - e_3\}} \prod_{\rho \in \Sigma(1)} X_{\rho}^{\langle m, v_{\rho} \rangle + 1}, \quad s_{\infty} = \prod_{\rho} X_{\rho}.$$

The exponents are nonnegative because $v_{\rho} \in \Delta$.

The centre is the Cartier divisor

$$H = V(s_0|_{D_{\Sigma}}) \subset D_{\Sigma}.$$

Resolve the pencil $[s_0 : s_{\infty}]$ by blowing up its boundary base curves (using their successive proper transforms), and denote the resulting threefold by $\tilde{V}_{\Sigma}$, with its map $f : \tilde{V}_{\Sigma} \rightarrow \mathbb{P}^1$. Write $\tilde{D}_{\Sigma} = f^{-1}(\infty)$ for the fibre at infinity. The expected mirror is

$$X^{\vee} = \tilde{V}_{\Sigma} \setminus \tilde{D}_{\Sigma}.$$

This is the $\mathbb{P}^3$ case of Harder's construction [6, Section 5.2 and Proposition 5.4]. Equivalently, on its finite pencil chart this is the affine modification obtained from the ideal (s_∞, s_0) ; the regular function on it is still denoted W .

To see the centres directly, restrict s_0 to D_ρ . Only the terms with $\langle m, v_\rho \rangle = -1$ survive. Up to an invertible monomial, their sum has three terms when v_ρ is a vertex of Δ , two terms when it is interior to an edge, and one term when it is interior to a facet. Thus the nonempty centres are exactly those on the four vertex divisors and the eighteen edge-interior divisors. This recovers the construction by sums of neighbouring monomials: each edge has integral length four, hence there are $4 + 6 \cdot 3 = 22$ relevant rays and $6 \cdot 4 = 24$ two-cones joining consecutive edge points. The facet-interior divisors have no base curve and are removed with the boundary. On a chart near one boundary component the centre has equation $a = 0$, $1 + u + v = 0$. Adjoining the quotient $b = (1 + u + v)/a$ gives

$$ab = 1 + u + v.$$

At a base-locus crossing corresponding to consecutive lattice points on an edge of Δ , the centre has ideal $(ab, 1 + u)$ and the surviving chart is

$$abc = 1 + u.$$

These are the mirrors of the positive and negative vertex models, respectively; their local HMS equivalences are established in [9]. Generic edge and regular charts reduce to the corresponding nodal-times-torus and torus models. Applying descent with the compatible local HMS identifications is expected to give

$$\mathrm{Perf} \mathcal{W}(\widehat{U}) \simeq \mathrm{Perf}(X^\vee) = D^b \mathrm{Coh}(X^\vee).$$

Related descent results. For established instances of this local-to-global approach, see Gammage–Shende's fanifold theorem and its compatibility with Viterbo restriction [5, Theorem 5.4 and Corollaries 5.8–5.9], and Morimura–Sibilla–Zhou's HMS theorem for complete intersections in algebraic tori, proved by gluing finite covers of products of higher dimensional pairs of pants [7]. In particular, their Corollary 4.12 gives the Viterbo fibre-product statement under the separatedness assumption of Lemma 4.10, verified for these covers in Appendix A.

2 The Abouzaid–Groman–Varolgüneş strategy for K3 surfaces

Let B be an integral affine two-sphere with twenty-four standard nodal singularities. As in Lecture 2, it determines a symplectic K3 surface

$$\pi : (M, \omega) \longrightarrow B$$

with a reference Lagrangian section R . We use the brane conventions of Lecture 3 and describe the grading more explicitly.

Away from the twenty-four critical points, the vertical tangent planes $V = \ker(d\pi)$ form an oriented Lagrangian polarization: the orientation of B and the symplectic pairing orient V . For a compatible almost complex structure J , an oriented frame of V determines the phase of a complex volume form on $TM = V \oplus JV$. This trivializes the phase bundle of the canonical line and hence defines the grading structure. An oriented change of frame has positive real determinant, so the phase is independent of that choice. The phase trivialization extends over the critical points: in a punctured four-ball its obstruction is a map $S^3 \rightarrow S^1$, which is

null-homotopic. This agrees, up to homotopy, with the grading defined by a nowhere vanishing complex two-form Ω_M : two trivializations differ by a map $M \rightarrow \mathbb{C}^*$, and $H^1(M; \mathbb{Z}) = 0$ for a K3 surface. We grade R using its transversality to the oriented vertical polarization; the self-Floer degree convention is the usual cohomological Morse grading.

2.1 The sheaf of functions defined by the reference section

For a compact subset $P \subset B$, put $K_P = \pi^{-1}(P)$ and define

$$\mathcal{A}(P) = \mathrm{Hom}_{\mathrm{Fuk}_{K_P}}(R, R) \otimes_{\Lambda_{\geq 0}} \Lambda, \quad \mathcal{F}(P) = H^0(\mathcal{A}(P)).$$

The supported category is taken in the ambient manifold M . Its products and restriction functors give a presheaf of Λ -algebras. First equip the compact subsets of B with the G -topology whose covers are finite families with the prescribed union. Their inverse images are weakly Poisson commuting, so the Mayer–Vietoris property of Lecture 3 applies [1].

The grading gives

$$H^j(\mathcal{A}(P)) = 0 \quad (j < 0).$$

Here is the index argument. Choose acceleration Hamiltonians pulled back from the base, constant on shrinking neighbourhoods of the nodal values, and then perturb to achieve nondegeneracy. In action–angle coordinates (q, θ) with $R = \{\theta = 0\}$, a chord for $h(q)$ with winding $m \in \mathbb{Z}^2$ corresponds to a critical point of $h(q) - \langle m, q \rangle$. Its degree is the Morse index, hence is 0, 1, or 2; the winding contributes no extra Maslov index in the vertical grading. Near a node the Hamiltonian is constant, and a small Morse perturbation on R gives the same range of degrees. These choices are cofinal for every compact P . Thus the Floer complexes have no negative-degree generators. In the completed telescope the only possible negative degree is the degree -1 relation term; its differential contains $1 - \text{shift}$, which is injective (read the components successively). Completion therefore introduces no negative cohomology. This argument does not assert the stronger degree-zero concentration needed later on small convex polygons. For two compact subsets, the resulting long exact sequence therefore gives

$$0 \longrightarrow \mathcal{F}(P_1 \cup P_2) \longrightarrow \mathcal{F}(P_1) \oplus \mathcal{F}(P_2) \xrightarrow{\mathrm{res}_1 - \mathrm{res}_2} \mathcal{F}(P_1 \cap P_2).$$

The zero on the left comes from $H^{-1}(\mathcal{A}(P_1 \cap P_2)) = 0$. Thus compatible local functions glue uniquely, and induction proves the sheaf condition for finite covers. This uses nonnegative grading; positive-degree cohomology is allowed.

Now restrict $\mathcal{F}$ to the G -topology of admissible polygons. These are locally rational polygons in the regular affine charts or the standard nodal model, together with their finite unions. A small polygon containing a node contains it in its interior and contains no other node. Covers are finite polygonal covers. In particular, we can work with sufficiently small convex admissible polygons around every point of B .

2.2 Constructing the analytic mirror

Conjecture 1 (The analytic mirror). *There exists a rigid analytic Calabi–Yau surface Y over Λ with a Stein continuous map $p : Y \rightarrow B$ such that*

$$p_* \mathcal{O}_Y \cong \mathcal{F}.$$

Here continuity means that inverse images preserve admissible opens and covers. The Stein condition means that every point of B has a convex admissible neighbourhood P such that $p^{-1}(Q)$ is affinoid for every convex admissible polygon $Q \subset P$.

This is a construction of Y once the required properties of $\mathcal{F}$ are established. On sufficiently small polygons, one must show that $\mathcal{F}(P)$ is a commutative affinoid algebra and that restriction to $Q \subset P$ identifies $\mathrm{Sp} \mathcal{F}(Q)$ with an affinoid domain in $\mathrm{Sp} \mathcal{F}(P)$. Compatibility with intersections and covers then allows the charts

$$Y_P = \mathrm{Sp} \mathcal{F}(P)$$

to glue. The resulting map to B identifies functions on $p^{-1}(P)$ with $\mathcal{F}(P)$. The desired local Floer calculation also gives

$$\mathcal{A}(P) \simeq \mathcal{F}(P)[0] \tag{1}$$

on these small convex polygons: degree-zero concentration makes the local A_∞ algebra formal. This is the local calculation used in the coherent-sheaf construction below.

Remark 2.1. For certain open bases, namely those of symplectic cluster manifolds, the corresponding closed-string construction was carried out by Groman–Varolgüneş [12, Theorem 1.1]. There the sheaf is $P \mapsto SH^0(\pi^{-1}(P); \Lambda)$, and the resulting analytic mirror has a canonical volume form. The expected open–closed comparison connects this sheaf with the reference-section formulation above.

2.3 Local generation and its consequences

The *local generation criterion at P* says that the supported open–closed map for the one-object category generated by R hits the unit:

$$1_{K_P} \in \mathrm{im} \left(HH_2(\mathcal{A}(P)) \xrightarrow{\mathrm{OC}_P} SH_M^0(K_P; \Lambda) \right). \tag{2}$$

Hochschild homology is taken in the completed analytic setting. We use homological grading: in real dimension four, degree two maps to degree-zero symplectic cohomology.

Conjecture 2 (Local generation). *For a sufficiently small convex admissible polygon around any point of B , the reference section R satisfies the local generation criterion.*

This includes polygons around nodes. The proposed verification starts with the corresponding Weinstein model, where the local section split-generates, and transports a Hochschild cycle hitting the unit to the supported theory. Shrinking the support gives room for the required energy estimates.

Pomerleano’s affine HMS criterion and the nodal model. For an affine log Calabi–Yau U , a *homological section* L has $HW^*(L, L)$ concentrated in degree zero and closed–open isomorphism $SH^0(U) \cong HW^0(L, L)$. If this algebra A is smooth, Pomerleano’s automatic generation criterion makes L split-generate and gives $\mathrm{Perf} \mathcal{W}(U) \simeq \mathrm{Perf}(A)$ [10, Proposition 1.3 and Corollary 1.4]. The mechanism is that Floer-theoretic finiteness makes the subcategory $\mathrm{Perf}(A)$ admissible; the smooth Calabi–Yau structure and connected HH^0 exclude a nontrivial complementary summand.

For the nodal surface model

$$U_{\mathrm{node}} = \{(u, v, z) \in \mathbb{C}^2 \times \mathbb{C}^* : uv = 1 + z\},$$

the reference section is a homological section, with smooth algebra

$$A_{\mathrm{node}} = k[x, y, z^{\pm 1}]/(xy - 1 - z) \cong k[x, y, (xy - 1)^{-1}].$$

This is the two-dimensional case of [10, Lemma 5.25 and Proposition 5.30], over a field of characteristic zero. It supplies the exact local HMS input. Passing its open–closed cycle to the supported theory in the compact M remains the comparison required by Conjecture 2.

Assuming these conjectures and the supported generation package, we note three consequences. The supported Cardy argument makes R split-generate locally. The open–closed comparison identifies local Hochschild invariants with the closed-string theory and supplies a smooth Calabi–Yau structure.

Smoothness of the analytic mirror. The local unit condition also implies that Y is smooth, provided the supported generation package includes the diagonal-generation argument. Fix a small convex polygon P and write $A_P = \mathcal{F}(P) = \mathcal{O}(Y_P)$. By (1), this commutative affinoid algebra is a model for $\mathcal{A}(P)$. The required implication is

$$1_{K_P} \in \text{im}(\text{OC}_P) \implies A_P \in \text{Perf}(A_P \hat{\otimes}_{\Lambda} A_P) \implies Y_P \text{ is smooth.}$$

For the first step, the unit-hitting cycle gives split-generation of the local diagonal by $R^- \times R$. Under the passage from product Lagrangians to bimodules, this product gives the free bimodule and the diagonal gives A_P . Finite cones and direct summands therefore make the diagonal bimodule perfect. In ordinary wrapped Floer theory this is Ganatra’s diagonal-generation theorem and its smoothness consequence [26, Theorems 1.5 and 1.2]. Here we require its supported version, compatible with the analytic completion. In particular, it must produce a perfect analytic diagonal; generation using only completed infinite constructions would not establish this conclusion.

The second step is a fact about affinoid geometry. Pulling the perfect diagonal back to $Y_P \times \{y\}$ gives a perfect residue-field module $\kappa(y)$ over $A_P \otimes_{\Lambda} \kappa(y)$. The residue extension is finite; the pullback has no higher Tor terms because the diagonal is flat over the second factor. Restriction of scalars therefore gives finite projective dimension over A_P , and hence over the local ring at y . The residue-field criterion for regularity applies. The same argument after finite extensions of Λ gives geometric regularity and thus smoothness. See [27, Proposition 5.3 and Corollary 5.4] for this analytic diagonal criterion. The tensor product above is the *completed analytic* tensor product. Applying the argument to the polygons in a cover proves that Y is smooth.

This uses more than the assertion that R split-generates locally: an algebra A generates $\text{Perf}(A)$ even when A is singular. The additional conclusion is perfection of the diagonal. Thus, with the intended supported diagonal-generation argument, smoothness is a consequence of the local unit condition, and is available before applying Hochschild–Kostant–Rosenberg to obtain a volume form.

An analytic volume form on Y . The inverse image of 1_{K_P} under the open–closed identification gives a distinguished Hochschild class. Under analytic Hochschild–Kostant–Rosenberg, this class becomes a two-form Ω_P on Y_P . Nondegeneracy of the Calabi–Yau structure makes it nowhere zero. Compatibility with restriction makes these local forms agree on overlaps, so they glue to

$$\Omega \in H^0(Y, \Omega_Y^2).$$

This gives the volume-form part of Conjecture 1.

A cohomologically fully faithful HMS functor. For a Lagrangian brane L , form the supported Yoneda modules

$$\mathcal{M}_P(L) = \text{Hom}_{\text{Fuk}_{K_P}}(R, L) \otimes_{\Lambda_{\geq 0}} \Lambda.$$

By (1) and local generation these are perfect modules over $\mathcal{F}(P) = \mathcal{O}(Y_P)$. Continuation identifies their restrictions, so they glue to a coherent complex $\mathcal{E}_L$. The resulting A_∞ functor is

$$\Phi : \mathrm{Fuk}(M; \Lambda) \longrightarrow \mathrm{Coh}^{dg}(Y), \quad L \longmapsto \mathcal{E}_L. \quad (3)$$

Here $\mathrm{Coh}^{dg}(Y)$ denotes a dg enhancement of the bounded derived category of coherent sheaves. In particular, $\Phi(R) = \mathcal{O}_Y$.

Choose a finite cover $B = \bigcup_i P_i$ by such small polygons, refining so the same local statements hold on their intersections. Write $K_I = \pi^{-1}(\bigcap_{i \in I} P_i)$ and $Y_I = p^{-1}(\bigcap_{i \in I} P_i)$. Then

$$\begin{aligned} CF_M^*(L, L'; \Lambda) &\simeq \mathrm{holim}_I CF_M^*(L, L'; K_I; \Lambda) \\ &\simeq \mathrm{holim}_I R\mathrm{Hom}_{\mathcal{O}(Y_I)}(\mathcal{M}_I(L), \mathcal{M}_I(L')) \\ &\simeq R\mathrm{Hom}_Y(\mathcal{E}_L, \mathcal{E}_{L'}). \end{aligned}$$

The first equivalence is support descent from Lecture 3, the second is local generation, and the third is analytic descent. This proves cohomological full faithfulness by descent for the same cover on both sides.

3 Relative Fukaya categories: the quartic and the quintic

This part describes ongoing joint work with Yuan Gao. We combine the quartic and quintic bases of Lecture 2, Sections 9.2 and 9.5, with the supported Floer theory and descent of Lecture 3, Sections 4–7. The relative construction retains a central fibre and a polarization. Its one-dimensional prototype is the direct calculation in Lecture 3, Section 9.5: allowing a triangle to cross the i th puncture changes $X_i Y_i = 0$ into $X_i Y_i = Q_i$. We now explain the proposed construction first for a quartic K3 and then for a quintic threefold.

3.1 The quartic K3 and its reference section

Let $M_2 \subset \mathbb{P}^3$ be a smooth quartic transverse to all coordinate strata, and put

$$D_i = M_2 \cap \{x_i = 0\}, \quad D = \bigcup_{i=0}^3 D_i, \quad R_2 = k[[Q_0, Q_1, Q_2, Q_3]].$$

Each D_i is a plane quartic, each pair meets in four points, and there are no triple intersections. Thus D is a simple normal crossings divisor of class $4H$, with $H = c_1(\mathcal{O}_{M_2}(1))$. Take $[\omega]$ proportional to H . The complement has an adapted primitive with positive residues, as in Lecture 2, Section 5.1.

Use the tetrahedral affine base of Lecture 2, Section 9.2:

$$B_2 = \partial \Delta_3, \quad \Delta_3 = \{a \in \mathbb{R}^3 : a_j \geq -1, \ a_1 + a_2 + a_3 \leq 1\}.$$

Its edges have integral length four, and there are four simple nodes on each of the six edges. Choose a coherent unimodular star triangulation of Δ_3 centred at 0, with height function $h(0) = 0$ and positive boundary heights. We work with a real patchworking representative of the symplectic pair, identified with the original pair by deformation and parallel transport. On the dense torus its equation is

$$-1 + \sum_{m \in \partial \Delta_3 \cap \mathbb{Z}^3} c_m q^{h(m)} z^m = 0, \quad c_m > 0. \quad (4)$$

Multiplying by $x_0x_1x_2x_3$ homogenizes this to degree four. Choose generic coefficients and a sufficiently small positive smooth parameter q . The geometric parameter q here is separate from the relative Floer parameters Q_i .

The compact positive-real component gives the reference Lagrangian $L_0 \cong S^2$. Indeed, in logarithmic coordinates $\rho \in \mathbb{R}^3$, its equation is $G(\rho) = 1$, where

$$G(\rho) = \sum_m c_m q^{h(m)} e^{\langle m, \rho \rangle}.$$

The exponents span $\mathbb{R}^3$ and contain 0 in the interior of their convex hull. Hence G is proper and strictly convex; for small q , $G(0) < 1$. The level set is the boundary of a bounded strictly convex domain, and is a two-sphere. Complex conjugation makes it Lagrangian. An adapted primitive satisfying $\iota^* \lambda = -\lambda$ vanishes on its fixed locus, so L_0 is exact.

The decomposition and the divisor. Write $\mathcal{T}$ for the triangulation and $\partial\mathcal{T}$ for its boundary triangulation. A maximal simplex $\text{Conv}(0, m_1, m_2, m_3)$ is unimodular. In the corresponding monomial coordinates the dominant terms give

$$-1 + w_1 + w_2 + w_3 = 0 \quad \text{in } (\mathbb{C}^*)^3.$$

This is the two-dimensional pair of pants, a projective plane with four lines in general position removed. *Tailoring* means cutting off the non-dominant monomials so its ends have product coordinates in which neighbouring pieces glue. The face opposite 0 lies in an original facet F_i of Δ_3 : the corresponding end is filled by D_i in the compactification. The other three ends are the interfaces with neighbouring pieces. For the compactification and product collars see [28, Proposition 4.6, Theorem 4 and Remark 5.2]; the monomial cutoffs are described in [8, Sections 1.2–1.3].

Each F_i has sixteen unimodular triangles. The dual trivalent graph records the pants decomposition of D_i with its twelve intersection points removed: it has sixteen vertices, eighteen bounded edges, and four ends on each side. Its first Betti number is three, as required for a plane quartic. Along an original tetrahedron edge, four ends from each adjacent facet pair up at the four points of $D_i \cap D_j$. Figure 3 shows this polyhedral compactification data on B_2 . In the finite-parameter geometric model, the graphs have tubular neighbourhoods. The polyhedral pants base is related to the sphere by the collapse described in [28, Section 4.2].

The RSTZ skeleton explicitly. Set $N = \mathbb{Z}^3$ and $T_N = \text{Hom}(N, \mathbb{R}/\mathbb{Z})$. For a boundary simplex σ set

$$T_\sigma = \{\theta \in T_N : \theta(m) = 0 \text{ for every vertex } m \text{ of } \sigma\}.$$

The Ruddat–Sibilla–Treumann–Zaslow skeleton is

$$\mathfrak{S}_{\mathcal{T}} = \bigcup_{\sigma \in \partial\mathcal{T}} \sigma \times T_\sigma \subset B_2 \times T_N.$$

Face inclusions glue these pieces: if $\sigma' \subset \sigma$, then $T_\sigma \subset T_{\sigma'}$. Unimodularity gives a point over an open triangle, a circle over an open edge, and a two-torus over a vertex. The locus $\theta = 0$ is the sphere B_2 , representing L_0 . Thus the skeleton consists of this reference sphere together with circle and torus pieces over the triangulation's lower-dimensional strata. The combinatorial construction and deformation retraction are in [29]; Zhou realizes it as a Liouville skeleton after tailoring and choosing an adapted potential [8, formula (0.2) and Main Theorem].

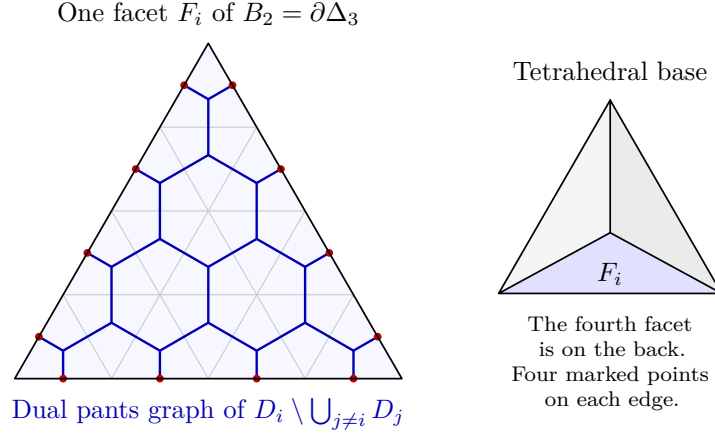


Figure 3: The polyhedral/tropical compactification diagram for the quartic. Each facet has sixteen unimodular triangles; coning these to the interior lattice point gives sixteen of the sixty-four two-dimensional pants pieces. The blue graph is dual to this subdivision and records the pants decomposition of the corresponding plane-quartic divisor. Its four ends on each side meet those from the adjacent facet at the red points. Globally there are $4 \cdot 12/2 = 24$ such points.

3.2 The quintic threefold

Now take a smooth quintic $M_3 \subset \mathbb{P}^4$, transverse to the coordinate strata, and set

$$D_i = M_3 \cap \{x_i = 0\}, \quad D = \bigcup_{i=0}^4 D_i, \quad R_3 = k[[Q_0, Q_1, Q_2, Q_3, Q_4]].$$

The components are quintic surfaces. Their pairwise intersections are plane quintic curves, every triple intersection has five points, and there are no intersections of four components. Thus D is simple normal crossings, with class $5H$. Again choose $[\omega]$ proportional to H and an adapted primitive on $M_3 \setminus D$.

The base is now the three-sphere of Lecture 2, Section 9.5:

$$B_3 = \partial\Delta_4, \quad \Delta_4 = \{a \in \mathbb{R}^4 : a_j \geq -1, \sum_j a_j \leq 1\}.$$

Its edges have integral length five. Use the subdivision and discriminant graph discussed there. Formula (4), with Δ_3 replaced by Δ_4 and four torus variables, homogenizes to a quintic. Again choose such a representative by deformation and parallel transport. The same convexity argument in $\mathbb{R}^4$ gives an exact positive-real reference section $L_0 \cong S^3$.

The compactified pair-of-pants pieces have complex dimension three. Along a generic discriminant edge the local model is the nodal surface model with one additional torus direction; at vertices one uses the positive and negative models of Lecture 2, Section 9.4. Thus the quintic requires relative calculations at both kinds of vertex, as well as compatibility along their overlaps. The same construction uses the three-dimensional pair of pants $-1 + w_1 + w_2 + w_3 + w_4 = 0$, whose compactification is $\mathbb{P}^3$ with five hyperplanes in general position. The RSTZ formula above now uses $N = \mathbb{Z}^4$ and B_3 ; its fibre over an open r -simplex is T^{3-r} , and $\theta = 0$ is the reference S^3 .

3.3 Relative operations and supported local algebras

In either example write $n = 2$ or 3 and work over R_n , with maximal ideal $\mathfrak{m} = (Q_0, \dots, Q_{n+1})$. A rigid polygon contributes

$$\epsilon(u) \prod_{i=0}^{n+1} Q_i^{u \cdot D_i}.$$

This is the relative theory introduced in Lecture 2, Sections 6.1–6.2, using the polygon operations of Lecture 1. Reduction modulo $\mathfrak{m}$ retains the polygons avoiding D and recovers the compact exact category of $M_n \setminus D$ on these objects. We use graded branes with unobstructed relative lifts, including bounding cochains when necessary; for the projective relative foundations see [13].

Choose a finite adapted cover by supports K_α in the tailored decomposition. The Hamiltonians defining restrictions preserve D and are functions of commuting local Hamiltonians. As in Lecture 3, one localizes at continuation elements generated by Hamiltonians negative on the support and then completes. In this relative setting, the local calculation must identify that completion with $\mathfrak{m}$ -adic completion and supply a coherent restriction diagram.

For a support K , put

$$\mathcal{A}_K = CF_{M_n, D; K}^*(L_0, L_0), \quad A_K = H^0(\mathcal{A}_K), \quad \mathcal{M}_K(L) = CF_{M_n, D; K}^*(L_0, L).$$

The required local statements are that $\mathcal{A}_K \simeq A_K[0]$, that A_K is commutative and flat over R_n , and that its restrictions give open embeddings of formal affine schemes. Local generation is the relative analogue of Section 2.3. With these inputs,

$$\mathfrak{Y}_K = \mathrm{Spf} A_K \quad \text{glue to} \quad \mathfrak{Y}_n,$$

and the modules glue to coherent complexes. The central charts $\mathrm{Spec}(A_K/\mathfrak{m})$ are computed by the exact puncture-avoiding operations. Terms of positive $\mathfrak{m}$ -order record the relative correction, just as $X_i Y_i = Q_i$ did in Lecture 3.

3.4 Monodromy, polarization, and the HMS functor

Fix a strictly convex multivalued integral piecewise-linear function φ for the chosen decomposition. Its local representatives differ by integral affine functions. After compatible smoothing, their Hamiltonian time-one maps agree on overlaps, since the extra integral affine Hamiltonians have period-one flows. With the grading lift and divisor compatibility fixed, write the resulting map as τ_φ and put

$$L_k = \tau_\varphi^k(L_0), \quad k \in \mathbb{Z}.$$

For the patchworking families, this is the monodromy construction described by tropical localization [15]. The HMS functor has already been specified by the modules $L \mapsto \mathcal{M}_K(L)$ and their descent. The calculation of its images is made on the *exact fibre*, obtained by setting all $Q_i = 0$. Write $Y_{n,0}$ for this central fibre and F_0 for the reduced functor. The monomial transitions arising from φ give an ample bundle $\overline{\mathcal{L}}$, and the calculation is

$$F_0(L_0) = \mathcal{O}_{Y_{n,0}}, \quad F_0(L_k) = \overline{\mathcal{L}}^{\otimes k}.$$

This uses the elementary local frames from Lecture 3, Sections 9.4–9.5, in the higher-dimensional pants models.

The images supplied by the relative functor are perfect. A perfect complex whose derived reduction is a line bundle is itself a line bundle on the formal neighbourhood. Put $\mathcal{L} =$

$F(L_1)$. For these toric central fibres, the resolution by toric strata computes $H^1(Y_{n,0}, \mathcal{O})$ as $H^1(S^n; k) = 0$. The exact sequence for a square-zero thickening then makes an existing line-bundle lift unique at each order [30, Tag 0C6Q]. Consequently

$$F(L_0) = \mathcal{O}_{\mathfrak{Y}_n}, \quad F(L_k) = \mathcal{L}^{\otimes k},$$

and $\mathcal{L}$ is relatively ample. The relative functor itself supplies these lifts; no separate extension of the transition functions is required.

Let $\mathcal{C}_n$ be the split-closed category generated by the L_k in the relative Fukaya category. Local generation and descent give

$$\begin{aligned} CF_{M_n, D}^*(L, L') &\simeq \operatorname{holim}_I CF_{M_n, D; K_I}^*(L, L') \\ &\simeq \operatorname{holim}_I R\operatorname{Hom}_{A_{K_I}}(\mathcal{M}_{K_I}(L), \mathcal{M}_{K_I}(L')) \\ &\simeq R\operatorname{Hom}_{\mathfrak{Y}_n}(\mathcal{E}_L, \mathcal{E}_{L'}). \end{aligned}$$

Equivalently, with compatible derived reduction and completed morphism complexes, full faithfulness can be checked on the exact fibre: the cone of each morphism map is derived complete and has zero reduction, hence vanishes by derived Nakayama [30, Tag 0G1U]. The image contains all powers of $\mathcal{L}$. These split-generate perfect complexes, so this yields

$$\mathcal{C}_n \simeq \operatorname{Perf}(\mathfrak{Y}_n).$$

For a proper noetherian formal model, the extended ample bundle algebraizes it to a projective R_n -scheme Y_n . Its section algebra is

$$S_n = \bigoplus_{k \geq 0} HF_{M_n, D}^0(L_0, L_k), \quad Y_n \cong \operatorname{Proj}_{R_n} S_n.$$

For $a \in HF^0(L_0, L_k)$ and $b \in HF^0(L_0, L_\ell)$, the product is $\mu^2(\tau_\varphi^k b, a)$. The mirror identification makes this the commutative multiplication of sections of powers of $\mathcal{L}$.

Bai–Haney give a related intrinsic construction of the mirror from fixed-point Floer cohomology of monodromy: see [14, Definitions 1.2 and 1.4] for their ring and relative Proj, along with a partial HMS functor in Theorem 1.6. Appendix A explains the direct comparison with their open-string algebra and the reconstruction of our polarized formal mirror.

Finally specialize the Q_i to Novikov elements of positive valuation matching the residues of the primitive. If the mirror is smooth and proper and the Floer-theoretic generation hypotheses hold, automatic generation upgrades the equivalence on $\mathcal{C}_n$ to

$$D^\pi \operatorname{Fuk}(M_n; \Lambda) \simeq \operatorname{Perf}(Y_{n, \Lambda}) = D^b \operatorname{Coh}(Y_{n, \Lambda}), \quad n = 2, 3.$$

Here we use the generation results of [18, 19, 20].

3.5 The Gross–Siebert comparison in the two examples

Fix the mirror toric log central fibre and its polarized affine data, including the gluing data. To pass from the quartic or quintic degeneration to the mirror data, use the discrete Legendre transform. For these toric hypersurfaces, the relevant construction is [16, Theorems 2.15, 3.16 and 3.25].

For the comparison, specialize $Q_i \mapsto c_i Q$, with $c_i \in k^*$, and use Q for the parameter denoted by t in [17]. Write $\mathcal{P}$ for the polyhedral decomposition on the Gross–Siebert side. To a cell ω associate the support K_ω over its dual cell $\omega^\vee$ on the symplectic side, using compatible small thickenings of the dual cells in our adapted cover. Thus $\omega \subseteq \omega'$ gives $K_{\omega'} \subseteq K_\omega$. Put

$$A_\omega = A_{K_\omega} \widehat{\otimes}_{R_n} k[[Q]], \quad A_\omega^{(N)} = A_\omega / (Q^{N+1}).$$

Here $A_{K_\omega} = HF_{M_n, D; K_\omega}^0(L_0, L_0)$ is the supported endomorphism algebra from Section 3.3.

The conjectural comparison of affine charts. The proposed identification, for every $N \geq 0$, is

$$\mathrm{Spec} A_{\omega}^{(N)} \cong V^N(\omega).$$

The schemes $V^N(\omega)$ are the flat affine deformations of the standard central-fibre charts $V(\omega)$ constructed in [17, Section 2.6, equation (2.14) and Proposition 2.38]. They assemble the chamber and stratum rings of that construction. For every incidence $\omega \subseteq \omega'$, support restriction should identify the following maps:

$$\begin{array}{ccc} \mathrm{Spec} A_{\omega'}^{(N)} & \xrightarrow{\text{support restriction}} & \mathrm{Spec} A_{\omega}^{(N)} \\ \cong \downarrow & & \downarrow \cong \\ V^N(\omega') & \xrightarrow{\Phi_{\omega\omega'}^N(s)} & V^N(\omega) \end{array}$$

The top arrow is induced contravariantly by $A_{\omega}^{(N)} \rightarrow A_{\omega'}^{(N)}$. Gross–Siebert prove that the bottom arrows are open embeddings [17, Proposition 2.39]; their colimit is the N th-order mirror. Hence compatible chart identifications identify the Floer-theoretic mirror with this colimit. Compatibility in N then identifies the formal mirrors, as in [17, Proposition 2.42]. This is the same comparison for the quartic K3 and the quintic threefold, with $n = 2$ and $n = 3$, respectively.

The first-order target for smoothness. A particularly useful goal is to establish this comparison only modulo Q^2 and show that this already forces the Floer-theoretic mirror to have smooth generic fibre. This would supply the smoothness input in Section 3.4 using only a first-order relative Floer calculation. Locally, such a calculation counts polygons avoiding D and polygons of total intersection number one with D .

The basic primitive model is

$$xy = Qf(z) + O(Q^2).$$

If f is a unit, or its zero locus in the transverse torus is smooth, the displayed first-order term forces smoothness after inverting Q . Indeed, for the relative Jacobian ideal J , the equation and its derivatives give $Qf, Q\partial_{z_j}f \in J + (Q^2)$. Thus $Q \in J + (Q^2)$, and $Q(1 - Qb) \in J$ for some b . Since $1 - Qb$ is a unit in the completed algebra, $Q \in J$. The same argument works for $x_1 \cdots x_r = Qf(z) + O(Q^2)$. The codimension-one Gross–Siebert model is $XY = f_{\rho}t^e$ [17, Lemma 2.34]; the primitive case $e = 1$ is visible at first order. The proposed first-order argument for the quartic and quintic must make the corresponding check at every central-fibre stratum, including the higher-codimension local models. Establishing this for those models is the smoothness target; the compatible identifications at every order above are the full mirror-comparison conjecture.

4 Sheaf quantization and the Tamarkin category

This last direction is ongoing work of Özel. Let $\pi : T^*Q \rightarrow Q$ be the cotangent projection, with its tautological Liouville form. We consider exact Lagrangian branes, including branes conical at infinity, together with chosen primitives. The target of the proposed quantization is the *Tamarkin category* [21, Section 2.2]:

$$\mathcal{T}(Q) = D(\mathrm{Sh}(Q \times \mathbb{R}_t; k)) / D_{\{\tau \leq 0\}}(Q \times \mathbb{R}_t; k).$$

The denominator consists of complexes whose microsupport is contained in $\{\tau \leq 0\}$. The variable t records action, and τ is its dual cotangent coordinate. Changing the additive constant of a primitive corresponds to translation in t . We use the action-filtered convention of Lecture 3, Section 4.2.

The extension of the supported theory. We extend the theory of Lecture 3 from compact supports to *closed supports which are conical at infinity*. The basic examples are $K^\sharp = \pi^{-1}(K)$ for closed subsets $K \subset Q$. Write

$$\mathcal{W}^{\mathrm{inf}}(T^*Q) := \mathrm{Fuk}_{T^*Q}(T^*Q).$$

The aim is a cohomologically fully faithful functor

$$\mathcal{W}^{\mathrm{inf}}(T^*Q) \longrightarrow \mathcal{T}(Q), \quad L \longmapsto \mathcal{S}_L,$$

with compatible filtrations on morphism spaces.

The analogue of the construction in Section 2. Use the zero section $R = Q \subset T^*Q$, with zero primitive, in place of the reference section. For compact $K \subset Q$, form

$$\mathcal{A}(K) = \mathrm{Hom}_{\mathrm{Fuk}_{T^*Q}(K^\sharp)}(R, R), \quad \mathcal{M}_L(K) = \mathrm{Hom}_{\mathrm{Fuk}_{T^*Q}(K^\sharp)}(R, L).$$

Products make $\mathcal{M}_L(K)$ a module over $\mathcal{A}(K)$, and continuation gives restriction maps when $K' \subset K$. These are complexes with their action data retained. We regard them in the sheaf model for filtered complexes on $\mathbb{R}_t$. The expected local zero-section calculation identifies the resulting *sheaf of algebras* with the convolution unit, represented by $k_{Q \times [0, \infty)}$. The modules then give the objects $\mathcal{S}_L$. To explain this precisely, we distinguish the description of a sheaf by its values on compact subsets of Q from its description by sections over open subsets.

The same sheaf on compact subsets and on open subsets. For example, fix L, L' and consider the compact-set assignment

$$\mathcal{F}_{L, L'}(K) = \mathrm{Hom}_{\mathrm{Fuk}_{T^*Q}(K^\sharp)}(L, L').$$

Once support descent and neighbourhood continuity have been established for the extended theory, this is already a *homotopy K -sheaf*. The same applies to $\mathcal{A}$ and $\mathcal{M}_L$. Thus no additional sheafification is needed to impose a gluing condition. The extension to these noncompact supports, with these properties, is part of the ongoing work described here.

Here is the precise compact-set formulation. Write $\mathcal{F}$ for any of these assignments, retaining its action-valued coefficients. The K -sheaf conditions are $\mathcal{F}(\emptyset) \simeq 0$, homotopy Mayer–Vietoris

$$\mathcal{F}(K_1 \cup K_2) \simeq \mathcal{F}(K_1) \underset{\mathcal{F}(K_1 \cap K_2)}{\overset{h}{\times}} \mathcal{F}(K_2),$$

and neighbourhood continuity

$$\operatorname{hocolim}_{\substack{K \subset \operatorname{int} K' \\ K' \text{ compact}}} \mathcal{F}(K') \xrightarrow{\simeq} \mathcal{F}(K),$$

where the maps follow shrinking compact neighbourhoods of K . The first displayed equivalence is the sheaf gluing condition on compact subsets; the second says that the value at K is recovered from arbitrarily small neighbourhoods. See [22, Appendix 1.A, Definition 1.A.2.5 and Remark 1.A.2.8].

Remark 4.1 (Neighbourhood continuity fails in general). The continuity property just stated is not a general property of supported Floer cohomology over the Novikov field Λ , with $\operatorname{val}(T) = 1$. Recall the annulus calculation from Lecture 3, Section 9.2; see also [1, Section 2.1, Remark 2.2]. In the area-one two-torus

$$M = (\mathbb{R}/\mathbb{Z})_p \times (\mathbb{R}/\mathbb{Z})_q, \quad \omega = dp \wedge dq, \quad \pi(p, q) = p,$$

take the section $L = \{q = 0\}$ and the closed annulus $K_I = \pi^{-1}(I)$, where $I = [a, b]$ is a proper interval in the base circle with a chosen lift $a < b < a + 1$. Over Λ , its supported endomorphism cohomology is concentrated in degree zero, with

$$HF_M^0(L, L; K_I; \Lambda) = A_I = \left\{ \sum_{m \in \mathbb{Z}} c_m z^m : \operatorname{val}(c_m) + \min(ma, mb) \longrightarrow +\infty \text{ as } |m| \longrightarrow \infty \right\}.$$

This is the algebra of analytic functions converging on the entire closed mirror annulus $a \leq \operatorname{val}(z) \leq b$. For small $\epsilon > 0$, put $I_\epsilon = [a - \epsilon, b + \epsilon]$. The maps $A_{I_\epsilon} \rightarrow A_I$ are restrictions of functions. Neighbourhood continuity would imply that every function in A_I extends to one of these strictly larger annuli.

Here is an explicit counterexample. Set $x = T^{-a}z$ and consider

$$f = \sum_{n \geq 1} T^n x^{n^2}.$$

On the closed annulus for I , one has $\operatorname{val}(x) \in [0, b - a]$, and the minimum valuation of the n th term is $n \rightarrow +\infty$. Thus the series converges uniformly there and defines an element of A_I . On the larger annulus for I_ϵ , the minimum valuation is

$$n - \epsilon n^2 \longrightarrow -\infty.$$

Consequently the terms do not even tend to zero there, for any $\epsilon > 0$. Uniqueness of Laurent expansions rules out an extension represented by a different series. Therefore the restriction map

$$\varinjlim_{\epsilon \downarrow 0} A_{I_\epsilon} \longrightarrow A_I$$

is not surjective. These enlarged annuli are cofinal among neighbourhoods of K_I , so this also detects the failure of the full neighbourhood-continuity map. Since the supported cohomology is concentrated in degree zero and filtered colimits over Λ are exact, the same example obstructs the corresponding uncompleted homotopy-colimit statement.

In terms of functions, the source consists of those converging on *some strictly larger annulus*; the target consists of all those converging on the given closed annulus. No completion of the direct limit is being taken here. Laurent polynomials belong to the source and are dense in the target for the annulus norm, so completing this union in that norm changes the answer. Thus continuity cannot be deduced from finite support descent alone; for the cotangent construction above it is a separate property to establish in the chosen action-valued coefficient category.

Under the stated K -sheaf hypotheses, since Q is locally compact and Hausdorff, these data equivalently specify a sheaf on its ordinary open-set topology. Its sections are

$$\mathcal{F}^{\text{op}}(U) = \operatorname{holim}_{\substack{K \subset U \\ K \text{ compact}}} \mathcal{F}(K).$$

The maps in this diagram are the restriction maps from larger compact subsets to smaller ones. Thus a section on U is a compatible system of compact-set sections, with the higher compatibilities retained. For an exhaustion $K_j \subset \operatorname{int} K_{j+1}$ with $\bigcup_j K_j = U$, this is the homotopy limit of the restriction tower $\mathcal{F}(K_{j+1}) \rightarrow \mathcal{F}(K_j)$.

Conversely, the original compact-set values are recovered by

$$\operatorname{hocolim}_{\substack{U \supset K \\ U \text{ open}}} \mathcal{F}^{\text{op}}(U) \xrightarrow{\sim} \mathcal{F}(K),$$

again using restrictions along shrinking neighbourhoods. These two constructions give an equivalence between the compact-set and open-set descriptions of a sheaf [23, Theorem 7.3.4.9 and Corollary 7.3.4.10]. In particular, the passage to open sets preserves all the compact-set data. It is not the operation of replacing a presheaf which fails descent by its associated sheaf. The notation $\mathcal{F}(K)$ here describes the value of the sheaf on K , recovered from neighbourhoods; it does not mean sections with compact support in K . Open-set sections use an inverse limit, whereas neighbourhood continuity uses a direct limit.

The action variable and the Tamarkin realization. There are two variables in this discussion. For each compact K , the value $\mathcal{F}(K)$ already retains its action data, expressed as a sheaf in the variable t . Descent and continuity in K organize these values into a sheaf over Q . All the limits above are taken in this category of action-valued complexes; the passage from compact to open subsets concerns only the Q variable.

In the Tamarkin realization, representatives on $U \times \mathbb{R}_t$ are obtained using the projector $F \mapsto F *_{\mathbb{R}} k_{[0, \infty)}$ [21, Propositions 2.1–2.2]. This projector acts in t and is a separate operation from the compact-set/open-set comparison in Q . Restriction in U commutes with it. Applied to the modules above, the construction therefore gives compatible objects of $\mathcal{T}(U)$ and hence $\mathcal{S}_L \in \mathcal{T}(Q)$.

Local comparison and full faithfulness. Let $\mathcal{H}_{L,L'}$ denote the sheaf on Q corresponding to the already existing supported morphism K -sheaf $\mathcal{F}_{L,L'}$. Explicitly,

$$\mathcal{H}_{L,L'}(U) = \operatorname{holim}_{\substack{K \subset U \\ K \text{ compact}}} \operatorname{Hom}_{\operatorname{Fuk}_{T^*Q}(K^\#)}(L, L').$$

As in Section 2, full faithfulness has a local and a descent step. The local comparison to establish identifies $\mathcal{H}_{L,L'}$ with the sheaf of action-valued morphisms between $\mathcal{S}_L$ and $\mathcal{S}_{L'}$. This is the analogue of the local-generation step in Section 2.3. Support descent and this comparison then give

$$\operatorname{Hom}_{\mathcal{W}_{\operatorname{inf}}^{\operatorname{act}}(T^*Q)}(L, L') \simeq R\Gamma(Q; \mathcal{H}_{L,L'}) \simeq R\operatorname{Hom}_{\mathcal{T}(Q)}^{\operatorname{act}}(\mathcal{S}_L, \mathcal{S}_{L'}),$$

where all three objects retain action. On the sheaf side, the superscript act denotes the translation-enhanced morphism object: its level- a complex is $R\operatorname{Hom}_{\mathcal{T}(Q)}(\mathcal{S}_L, T_{a*}\mathcal{S}_{L'})$, with the matching action convention on the Floer side. Level zero gives cohomological full faithfulness. The supported Yoneda operations supply the A_∞ functor. The extension of the supported theory and the local comparison are part of the ongoing work.

A Supported affine charts and the open-string section algebra

This appendix explains the comparison with the open-string construction of Bai–Haney. The point is that the mirror built in Section 3 by gluing supported affine charts can also be recovered from the homogeneous Floer algebra of the monodromy orbit of the reference section. We keep the relative divisor throughout, so the comparison takes place over the formal coefficient ring. The geometric starting point is to normalize the convex potential φ to vanish on a chosen maximal polytope. Multiples of this normalized representative supply the acceleration data for that support. This relates the supported construction directly to the monodromy orbit; the section-ring and Proj arguments below express the consequences of this comparison.

A.1 The relative data and the descent inputs

Let M be the quartic K3 or quintic threefold of Section 3, with the specified divisor $D = \bigcup_{i=1}^r D_i$. Reindexing its components, put

$$R = k[[Q_1, \dots, Q_r]], \quad \mathfrak{m} = (Q_1, \dots, Q_r).$$

The one-parameter version is obtained by a fixed positive specialization of these variables. Every polygon, continuation map and higher compatibility map is weighted by its intersections with D :

$$\epsilon(u) \prod_i Q_i^{u \cdot D_i}.$$

We use divisor-preserving Hamiltonians and compatible relative brane data, including bounding cochains when required. The reduction modulo $\mathfrak{m}$ is the exact theory in $M \setminus D$.

For the adapted supports K and their intersections, write

$$\mathcal{A}_K = CF_{M,D;K}^*(L_0, L_0), \quad A_K = H^0(\mathcal{A}_K), \quad \mathcal{M}_K(L) = CF_{M,D;K}^*(L_0, L).$$

We use the local inputs of Section 3.3: the algebras are concentrated in degree zero, commutative, flat and noetherian over R , and are $\mathfrak{m}$ -adically complete; their restriction maps give open immersions of formal affine schemes. Assume that the charts $\mathrm{Spf} A_K$ glue to a proper noetherian formal scheme $\mathfrak{Y}$ over $\mathrm{Spf} R$. The relative supported modules then define a functor

$$F : \mathcal{C} \longrightarrow \mathrm{Perf}(\mathfrak{Y}), \quad F(L)|_{\mathrm{Spf} A_K} = \mathcal{M}_K(L),$$

where $\mathcal{C}$ is the split-closed subcategory generated by the monodromy iterates of L_0 . Local generation and descent are assumed to give the full faithfulness of this functor, as in Section 3.4. Its normalization is $F(L_0) = \mathcal{O}_{\mathfrak{Y}}$, since locally $\mathcal{M}_K(L_0)$ is the regular A_K -module.

All of this is relative over R . In using the support construction, we require its continuation and completion procedures to be compatible with reduction modulo $\mathfrak{m}^{N+1}$. This is the completion compatibility stipulated in Section 3.3.

A.2 Normalized potentials and acceleration on a maximal polytope

Let P be any maximal-dimensional polytope of the decomposition of B . The function φ is affine on P . Subtracting the affine extension ℓ_P of this restriction gives a representative

$$\varphi_P = \varphi - \ell_P, \quad \varphi_P|_P = 0.$$

Thus every maximal polytope has a representative of the same multivalued function vanishing on it. In the affine development where this convex representative is used, convexity gives

$\varphi_P \geq 0$; strict convexity across the facets makes it positive away from P . The supporting affine function ℓ_P is exactly what must be subtracted to arrange this.

This normalization is adapted to the support K_P lying over P . Before smoothing, the acceleration profiles are simply

$$h_{P,j} = j\varphi_P - \epsilon_j, \quad \epsilon_j \downarrow 0.$$

On P they increase to zero from below. Away from P they tend to $+\infty$, locally uniformly on compact subsets disjoint from P . After pulling back to the symplectic local model, smoothing the corners, and making admissible perturbations, they give an acceleration datum $H_{P,j}$ with

$$H_{P,j}|_{K_P} < 0, \quad H_{P,j}|_{K_P} \longrightarrow 0, \quad H_{P,j} \longrightarrow +\infty \text{ off } K_P.$$

The smoothing is chosen with shrinking collars and errors tending to zero on K_P ; the small downward shifts leave room to impose strict monotonicity and transversality. These are the acceleration conditions of Lecture 3 and [1, Section 7]. The supported complex is computed by the completed continuation telescope of the resulting Floer complexes.

Since φ is multivalued, this uses normalized branches on suitable cut or developed affine domains. The construction does not require a globally single-valued convex function on the compact base B . Compatible realization across the cuts and the discriminant, preservation of D , and the locality comparison for the continuation diagrams are the technical parts of implementing these profiles in the relative supported theory.

The relation to monodromy is now visible. The affine function ℓ_P has integral slope, so its Hamiltonian flow is period one in the torus directions. Subtracting $j\ell_P$ therefore leaves the integer time-one map unchanged, with the sign convention fixed below. Consequently the normalized profiles and the monodromy powers have the same local integer-time maps. Before perturbation, the normalized map is the identity in the interior of P ; continuation extends this identification across the smoothing collars to each monodromy iterate. Keeping the additive constants is essential: they fix the actions and hence the coefficient normalization. This common continuation construction is the geometric basis of the comparison, including its relative weights and completion.

A.3 Monodromy and the polarization

Write $\tau = \tau_\varphi$ for the graded monodromy action determined by the multivalued integral piecewise-linear function φ , and put $L_d = \tau^d L_0$. In an action-angle chart the local model, after smoothing the potential and fixing the sign convention, is

$$(p, \theta) \longmapsto (p, \theta + d\varphi(p)).$$

Use the representatives φ_P and their acceleration data from Section A.2. The resulting local continuation classes give frames

$$\beta_P^{(d)} \in HF_{M,D;K_P}^0(L_0, L_d), \quad \beta_P^{(0)} = 1,$$

for the rank-one supported modules of L_d . Iteration and gluing of continuation trajectories give, on cohomology,

$$\beta_P^{(d+e)} = \mu^2(\tau^d \beta_P^{(e)}, \beta_P^{(d)}),$$

where the local identifications of the supported diagrams are understood. At chain level these equalities are supplied by the corresponding coherent continuation homotopies. Comparing

the normalizations for two polytopes on a common support gives the transition units of the mirror line bundle $\mathcal{L}$.

The compatibility needed for the comparison is a coherent identification on the monodromy-orbit category

$$F \circ \tau \simeq (- \otimes \mathcal{L}) \circ F. \quad (5)$$

Coherence means that the identifications for successive powers agree with iteration and with composition of morphisms. This is the morphism-level content of the normalized acceleration and continuation calculation just described. If its overlap units are $g_{\alpha\beta}$, iteration gives $g_{\alpha\beta}^d$ for L_d , and the product identifications multiply these units. Thus the same calculation which produces the line bundle must also identify the monodromy action on morphisms with tensoring by that bundle.

In particular, (5) gives compatible identifications

$$F(L_d) \simeq \mathcal{L}^{\otimes d}.$$

Objectwise identifications alone would not establish the assertion about the homogeneous multiplication below. Equation (5) is the conclusion required from the coherent local comparison; the compatible smoothing, relative Floer data and locality checks above are its geometric inputs. The bundle $\mathcal{L}$ is assumed to lift the ample bundle on the central fibre constructed from φ in Section 3.4.

A.4 Recovering the polarized formal mirror

Consider the homogeneous open-string algebra

$$S = \bigoplus_{d \geq 0} S_d, \quad S_d = HF_{M,D}^0(L_0, L_d), \quad a \star b = \mu^2(\tau^d b, a), \quad (6)$$

for $a \in S_d$ and $b \in S_e$. The integer d is the homogeneous degree, distinct from Floer cohomological degree and from $\mathfrak{m}$ -adic order. Completions are taken degree by degree in the coefficients; we keep the direct sum over d in forming S .

Full faithfulness identifies the graded pieces by

$$S_d \xrightarrow{F} \mathrm{Hom}_{\mathrm{Perf}(\mathfrak{Y})}(\mathcal{O}_{\mathfrak{Y}}, \mathcal{L}^{\otimes d}) = H^0(\mathfrak{Y}, \mathcal{L}^{\otimes d}).$$

To identify the product, view $F(a)$ as a map $\mathcal{O} \rightarrow \mathcal{L}^d$ and $F(b)$ as a map $\mathcal{O} \rightarrow \mathcal{L}^e$. Equation (5) identifies $F(\tau^d b)$ with the map $\mathcal{L}^d \rightarrow \mathcal{L}^{d+e}$ obtained by tensoring $F(b)$ with $\mathcal{L}^d$. Hence

$$F(a \star b) = (\mathrm{id}_{\mathcal{L}^d} \otimes F(b)) \circ F(a).$$

This is ordinary multiplication of sections. We have therefore proved an isomorphism of graded R -algebras

$$S \cong \bigoplus_{d \geq 0} H^0(\mathfrak{Y}, \mathcal{L}^{\otimes d}). \quad (7)$$

In particular, the multiplication in (6) is commutative under these hypotheses.

Here are the algebraization and reconstruction steps. Properness and the compatible ample line bundle allow us to apply Grothendieck's algebraization theorem: there is a projective R -scheme Y with an ample line bundle $\mathcal{L}_Y$ whose completion is $(\mathfrak{Y}, \mathcal{L})$ [30, Tag 089A]. Formal functions [30, Tag 087U] give

$$H^0(Y, \mathcal{L}_Y^{\otimes d}) \cong \varprojlim_N H^0(Y_N, \mathcal{L}_Y^{\otimes d}|_{Y_N}) = H^0(\mathfrak{Y}, \mathcal{L}^{\otimes d}),$$

where $Y_N = Y \times_R R/\mathfrak{m}^{N+1}$. The section modules are finite over the complete ring R , so no further completion of an individual graded piece is necessary. Thus S is the full section ring of $(Y, \mathcal{L}_Y)$; it is a finitely generated R -algebra [30, Tag 0B5T].

An ample line bundle reconstructs a proper scheme from its full section ring. The canonical map is an isomorphism

$$Y \xrightarrow{\sim} \mathrm{Proj}_R S, \quad \mathcal{L}_Y \cong \mathcal{O}_{\mathrm{Proj}_R S}(1).$$

The second statement uses the full section ring: its twisting sheaf $\mathcal{O}(1)$ is invertible and recovers the original line bundle. Generation of S in degree one is unnecessary. See [30, Tags 0C6J and 01QI]. Taking formal completions gives

$$(\mathfrak{Y}, \mathcal{L}) \cong (\widehat{\mathrm{Proj}_R S}, \widehat{\mathcal{O}(1)}). \quad (8)$$

The completion is along the closed fibre defined by $\mathfrak{m}$. This is the precise meaning of recovering the same *polarized formal mirror*.

No split-generation statement for the whole relative Fukaya category is needed for (8). Full faithfulness on the monodromy orbit, its compatibility with tensoring, and the stated properness and ampleness inputs suffice. Likewise, the argument does not require smoothness of the generic fibre.

A.5 Seeing the supported charts inside Proj

There is a direct continuation interpretation of homogeneous localization. Suppose the normalized continuation frame over P is the restriction of a closed homogeneous class $s_P \in S_r$, allowing a positive power and a refinement of the support if necessary. The comparison to construct identifies multiplication by s_P with the acceleration maps coming from multiples of φ_P :

$$S_0 \xrightarrow{*s_P} S_r \xrightarrow{*s_P} S_{2r} \xrightarrow{*s_P} \cdots$$

At stage j , a class $a \in S_{jr}$ represents a/s_P^j . Algebraically,

$$\varinjlim_j S_{jr} \cong (S[s_P^{-1}])_0.$$

The normalized acceleration calculation compares the completed continuation telescope with the supported complex. Under the degree-zero and completion compatibility inputs above, it therefore gives

$$A_{K_P} \cong (\widehat{S[s_P^{-1}]})_0,$$

with $\mathfrak{m}$ -adic completion. Constructing the global closed class s_P and checking this continuation comparison are part of the Floer calculation; the normalization alone supplies the local frame. Thus the geometric argument uses the normalized acceleration data to produce the chart comparison. The local-frame construction in the preceding subsections already suffices for the descent and section-ring argument, without requiring such a global class in degree one.

For completeness, the general algebraic chart description is as follows. Choose a homogeneous element $s \in S_d$, $d > 0$, whose corresponding section defines an affine nonvanishing locus $Y_s \subset Y$. Such loci form an affine cover by ampleness. Set

$$S_{(s)} = (S[s^{-1}])_0.$$

Then the section-ring construction identifies

$$\Gamma(Y_s, \mathcal{O}_Y) = S_{(s)}, \quad \widehat{Y}_s = \mathrm{Spf} \widehat{S_{(s)}}, \quad \widehat{S_{(s)}} = \varprojlim_N S_{(s)}/\mathfrak{m}^{N+1} S_{(s)}.$$

After refining the supported affine cover and this principal-open cover to a common affine cover, these are precisely the local rings of the same formal scheme. If a chosen supported chart $\mathrm{Spf} A_K$ is identified with $\widehat{Y}_s$, the map is explicitly

$$\widehat{S}_{(s)} \longrightarrow A_K, \quad \frac{a}{s^\ell} \longmapsto \frac{\mathrm{res}_K(a)}{\mathrm{res}_K(s)^{\otimes \ell}}, \quad a \in S_{\ell d}. \quad (9)$$

Here $\mathrm{res}_K(s)$ is a nowhere-vanishing section of $\mathcal{L}^d$ on this chart; dividing by it means using that section as a local frame. The numerator and denominator are sections of the same line bundle, so their ratio lies in A_K . The multiplication compatibility proved above makes this a ring homomorphism. The identification with Y_s makes it an isomorphism.

This description does not require every original support to correspond to a single principal open. Refinement is sufficient. On common subcharts, (9) commutes with support restriction because both the numerator and the chosen frame restrict. It therefore identifies the gluing maps as well as the affine rings.

Equivalently, the completed chart ring is

$$\widehat{S}_{(s)} = \varprojlim_N \left((S/\mathfrak{m}^{N+1}S)[s_N^{-1}] \right)_0,$$

where s_N is the image of s . This identity follows by commuting localization and quotient. It does not assume that $S_d/\mathfrak{m}^{N+1}S_d$ equals $H^0(Y_N, \mathcal{L}_Y^d|_{Y_N})$ in every degree. For a nonempty chart of the formal mirror, s must have a nonempty nonvanishing locus on the central fibre. Inverting a section divisible by the one-parameter variable q would give an empty chart on that fibre.

A.6 The direct comparison with Bai–Haney

Bai–Haney’s open-string construction uses the category generated by the iterates $\{\Phi^d L\}_{d \in \mathbb{Z}}$ of a tropical Lagrangian section. Its nonnegative homogeneous, cohomological-degree-zero part is the algebra

$$\bigoplus_{d \geq 0} HF^0(L, \Phi^d L), \quad a \star b = \mu^2(\Phi^d b, a),$$

with the monodromy-twisted operations of [14, Section 1.3].

For a direct relative comparison, take $L = L_0$ and use our divisor D , coefficient variables and brane data. Identify the graded monodromy action Φ with τ_φ on this relative category, compatibly with iteration. With these choices, its open-string algebra is (6): both count the same relative polygons with the same intersection weights. The representatives φ_P vanishing on maximal polytopes identify the local monodromy data with the support-acceleration data, as in Section A.2. Compatible continuation and polygon maps then identify the products as well as the individual Floer groups.

Thus the relative version of their open-string algebra, formed with our relative data, recovers the formal mirror by (8). The supported construction supplies its affine charts and their restrictions; the homogeneous algebra supplies its projective presentation and polarization. No comparison with closed-string cappings is needed for this statement.

Their paper defines its intrinsic mirror using a fixed-point Floer ring, and relates that ring to the open-string algebra by twisted closed–open; see [14, Proposition 1.19 and Lemma 5.11]. That comparison is carried out over Novikov coefficients. Identifying their particular fixed-point formal model with the model above is a further assertion about the coefficient normalization. Their Section 1.4 discusses the dependence of relative formal models on the divisor. The direct open-string comparison here fixes the divisor from the outset.

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